

Tutorial-7

Q1

Solution

$$X_L = X_C = 40 \Omega$$

$$I = \frac{E}{R} = \frac{20 \text{ mV}}{2 \Omega} = 10 \text{ mA}$$

$$V_R = I \times R = 10 \text{ mA} \times 2 = 20 \text{ mV}$$

$$V_L = I \times X_L = 10 \times 40 = 400 \text{ mV}$$

$$V_C = 10 \times 40 = 400 \text{ mV}$$

$$Q = \frac{X_L}{R} = \frac{40}{2} = 20$$

$$\begin{aligned} X_L &= 2\pi fL \\ &= 2 \times \pi \times 5 \times 10^3 \times L \\ &= 40 \end{aligned}$$

$$\begin{aligned} \text{or, } L &= \frac{40}{2\pi \times 5 \times 10^3} \\ &= 1.273 \text{ mH} \end{aligned}$$

$$X_C = 40 = \frac{1}{2\pi \times 5 \times 10^3 \times C}$$

$$\therefore C = 0.79577 \mu\text{F}$$

$$\Delta f = \frac{f_r}{Q} = \frac{5 \times 10^3}{20} = 250 \text{ Hz}$$

$$f_1 = f_r - \frac{\Delta f}{2} = 5000 - \frac{250}{2} = 4875 \text{ Hz}$$

$$f_2 = f_r + \frac{\Delta f}{2} = 5000 + \frac{250}{2} = 5125 \text{ Hz}$$

Q2

$$\Delta f = \frac{f_r}{Q} = \frac{6000}{15} = 400$$

$$f_1 = f_r - \Delta f/2 = 5800 \text{ Hz}$$

$$f_2 = f_r + \Delta f/2 = 6200 \text{ Hz}$$

$$Q = \frac{X_L}{R}$$

$$X_L = 75 \times 3$$

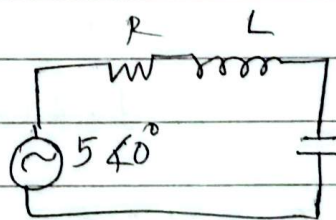
$$\therefore X_L = 45 \Omega = X_C$$

$$\begin{aligned} \text{Current at half power point} &= \frac{1}{2} I_0^2 R \\ &= 375 \text{ mW} \end{aligned}$$

Q3

At resonance

$$R = \frac{E}{I} = \frac{5}{500 \times 10^{-3}} = 10 \Omega$$



$$Q = \frac{f_r}{\Delta f} = \frac{8400}{120} = 70$$

$$Q = \frac{X_L}{R} \quad \text{or,} \quad X_L = 70 \times 10 = 700 \Omega = X_C$$

$$L = \frac{X_L}{2\pi f_r} = \frac{700}{2\pi \times 8400} = 13.26 \text{ mH}$$

$$C = \frac{1}{2\pi f_r \times C} = \frac{1}{2\pi \times 8400 \times 700} = 27.067 \text{ nF}$$

Cutoff frequencies are,

$$f_2 = 8400 + \frac{120}{2} = 8460 \text{ Hz}$$

$$f_1 = 8400 - \frac{120}{2} = 8340 \text{ Hz}$$

Q4) At generator,

$$V_{LG} = 208 \text{ V}$$

$$V_{pG} = 120.088 \text{ V}$$

At load, $V_{pL} = V_{pG} = 120.088 \text{ V}$

$$I_{pL} = \frac{V_{pL}}{1.0} = 12.01 \text{ A}$$

$$I_{LL} = I_{pL} = 12.01 \text{ for Y-connection.}$$

Q5
Soln:

$$V_L = 208 \text{ V}, V_p = 120.088 \text{ V}, \cos \theta = 0.6, \theta = 53.13^\circ$$

$$I_p = \frac{P}{3V_p \cos \theta} = \frac{1200}{3 \times 120.088 \times 0.6} = 5.551 \angle 53.13^\circ$$

$$\therefore Z_p = \frac{V_p}{I_p} = \frac{120.088}{5.551 \angle 53.13^\circ} = 21.633 \angle -53.13^\circ$$

$$= (12.979 - j17.306) \Omega$$

Q(6) $V_L = 220V$

$$V_p = \frac{220}{\sqrt{3}} = 127.017V \angle 0^\circ$$

Also,

$$I_p = \frac{V_p}{Z_p} = \frac{127.017}{10 - j10} = \frac{127.017}{14.142 \angle -45^\circ} = 8.98 \angle 45^\circ$$

$$\therefore I_L = I_p = 8.98 \text{ for } Y\text{-connection}$$

87) Soln:

$$\begin{aligned} E_{AB} &= 22kV \angle 0^\circ & E_{AN} &= \frac{22}{\sqrt{3}} \angle -30^\circ = 12.7kV \angle -30^\circ \\ E_{BC} &= 22kV \angle -120^\circ & E_{BN} &= 12.7kV \angle -150^\circ \\ E_{CA} &= 22kV \angle +120^\circ & E_{CN} &= 12.7kV \angle 90^\circ \end{aligned}$$

$$\begin{aligned} I_{AN} &= \frac{E_{AN}}{Z} = \frac{12.7 \angle -30^\circ}{0.43 + j1.04} = \frac{12.7 \angle -30^\circ}{1.1253 \angle 67.536^\circ} \\ &= 11.285 \angle -97.536^\circ A \end{aligned}$$

By,

$$I_{BN} = 11.285 \angle -217.536^\circ A$$

$$I_{CN} = 11.285 \angle 22.46^\circ A$$

As, Y-Y system is considered,

$$I_{Aa} = I_{AN} = 11.285 \angle -97.536^\circ A$$

$$I_{Bb} = I_{BN} = 11.285 \angle -217.536^\circ A$$

$$I_{Cc} = I_{CN} = 11.285 \angle 22.46^\circ A$$

$$V_{an} = I_{AN} \times Z_{an}$$

$$= 11.285 \angle -97.536^\circ \times (0.4 + j1) \times 10^3$$

$$= 11.285 \angle -97.536^\circ \times 1.077 \angle 68.198^\circ$$

$$= 1215.94 \angle -29.34^\circ$$

lly,

$$V_{bn} = 12153.94 \angle -149.34^\circ$$

$$V_{cn} = 12153.94 \angle 90.66^\circ$$

Q.6

$$I_{ab} = \frac{V_{ab}}{Z_{ab}} = \frac{16 \angle 0^\circ}{1.044 \angle 73.3^\circ} = 15.325 \angle -73.3^\circ$$

$$I_{bc} = \frac{V_{bc}}{Z_{bc}} = 15.325 \angle -193.3^\circ$$

$$I_{ca} = \frac{V_{ca}}{Z_{ca}} = 15.325 \angle 46.7^\circ$$

now,

$$I_{Aa} = I_{ab} - I_{ca} = 15.325 \angle -73.3^\circ - 15.325 \angle 46.7^\circ$$

$$= 4.4038 - j14.678 - 10.51 - j11.153$$

$$= -6.1062 - j25.831 = 26.54 \text{ A } \angle -103.3^\circ$$

Similarly,

$$I_{Bb} = I_{bc} - I_{ab} = -14.914 + j3.525 - 4.4 + j14.678$$

$$= -19.314 + j18.203$$

$$= 26.54 \text{ A } \angle 136.69^\circ$$

$$I_{Cc} = I_{ca} - I_{bc}$$

$$= 10.51 + j11.153 + 14.914 - j3.525$$

$$= 25.424 + j7.628$$

$$= 26.54 \angle 16.7^\circ$$

$$\begin{aligned}
 E_{AB} &= I_{Aa} (10 + j20) + V_{ab} - I_{Bb} (10 + j20) \\
 &= 26.54 \angle -103.3^\circ \times 22.36 \angle 63.435^\circ + 16000 - 26.54 \angle 136.69^\circ \\
 &\quad \times 22.36 \angle 63.435^\circ \\
 &= 455.4946 - j380.38 + 16000 + 557.201 + j204.182 \\
 &= 17012.6956 - j176.198 \\
 &= 17013.6 \angle -0.59^\circ
 \end{aligned}$$

$$E_{BC} = 17013.6 \angle -120.59^\circ$$

$$E_{CA} = 17013.6 \angle 119.41^\circ$$

Q9) $V_L = 16000 \text{ V}$, $V_{ph} = 16000/\sqrt{3} = 9237.6 \text{ V}$, $I_L = 80 \text{ A}$

power delivered by the source,

$$P = 1200000 + 3 \times (80)^2 \times 4 = 1276.8 \text{ kW}$$

$$P = 3V_p I_p \cos \phi \Rightarrow \cos \phi = \frac{1276800}{3 \times 9237.6 \times 80} = 0.576 \text{ lagging}$$

$$\therefore \phi = -54.83^\circ$$

$$\therefore I_{Aa} = 80 \angle -54.83^\circ$$

$$\begin{aligned}
 V_{an} &= E_{AN} - I_{Aa} \times (4 + j20) \\
 &= 9237.6 \angle 0^\circ - 80 \angle -54.83^\circ \times 20.396 \angle 78.69^\circ \\
 &= 9237.6 - 1492.231 - j660.02 \\
 &= 7773.44 \text{ V} \angle -4.87^\circ
 \end{aligned}$$

$$\therefore Z = \frac{V_{an}}{I_{an}} = \frac{7773.44 \angle -4.87^\circ}{80 \angle -54.83^\circ} = (62.51 + j74.39) \Omega$$

$$\cos \phi = \cos (49.96^\circ) = 0.643$$

$$\eta = \frac{1200}{1276.8} = 93.98\%$$